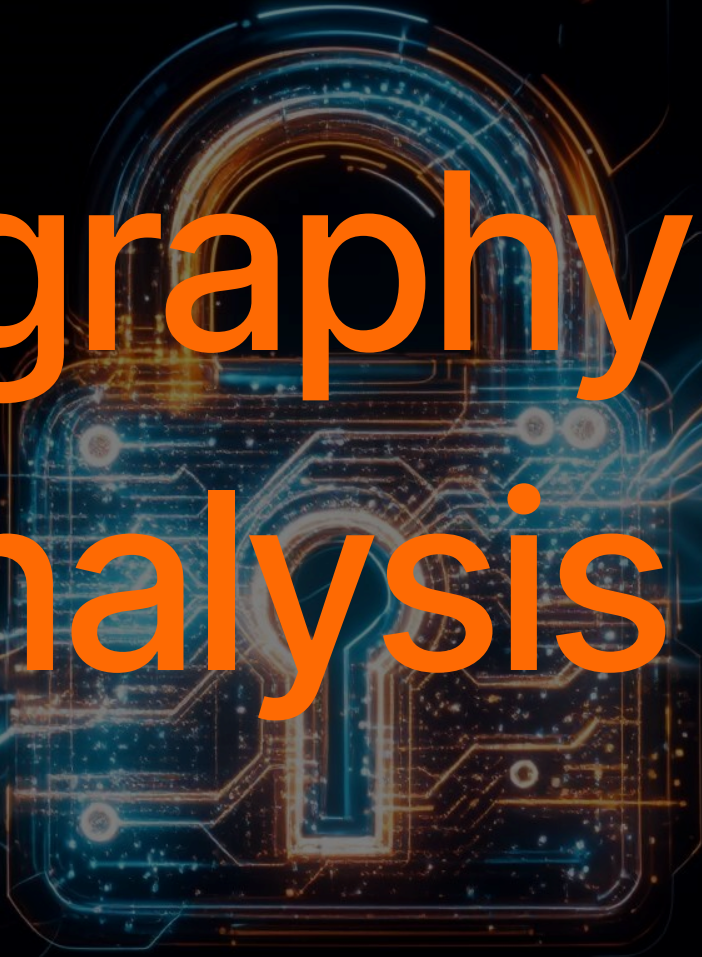


Cryptography & Cryptanalysis Toolkit



Caesar Cipher, Vigenère Cipher & Frequency Analysis

Jatin Mehta

React • FastAPI • Python • PostgreSQL

Made with GAMMA

Project Overview

This project is a web-based cryptography toolkit designed to demonstrate classical encryption techniques and basic cryptanalysis. It provides an interactive interface for exploring how classical ciphers work — and how they can be broken.



- Caesar Cipher**
Encryption, Decryption & Brute Force
- Vigenère Cipher**
Polyalphabetic encryption & decryption
- Frequency Analysis**
Letter frequency with visual bars

**Cryptography Toolkit**
Encryption, Decryption & Cryptanalysis

● API Connected

Cryptography & Cryptanalysis

Explore classical encryption techniques and analyze encrypted text using an interactive toolkit.

 Caesar Cipher

 Vigenère Cipher

 Frequency Analysis

 History

Caesar Cipher

Shift letters by a fixed number to encrypt or decrypt your message.

Input Text

Enter your message here...

CLASSICAL CIPHER

Made with GAMMA

Why This Project?

Classical ciphers provide a foundation for understanding how encryption and cryptanalysis work. This toolkit bridges programming, mathematics, and cybersecurity in a hands-on way.

“

"Understanding how simple ciphers can be built — and broken — gives you genuine insight into the principles behind modern cryptography."

— **Project Motivation**

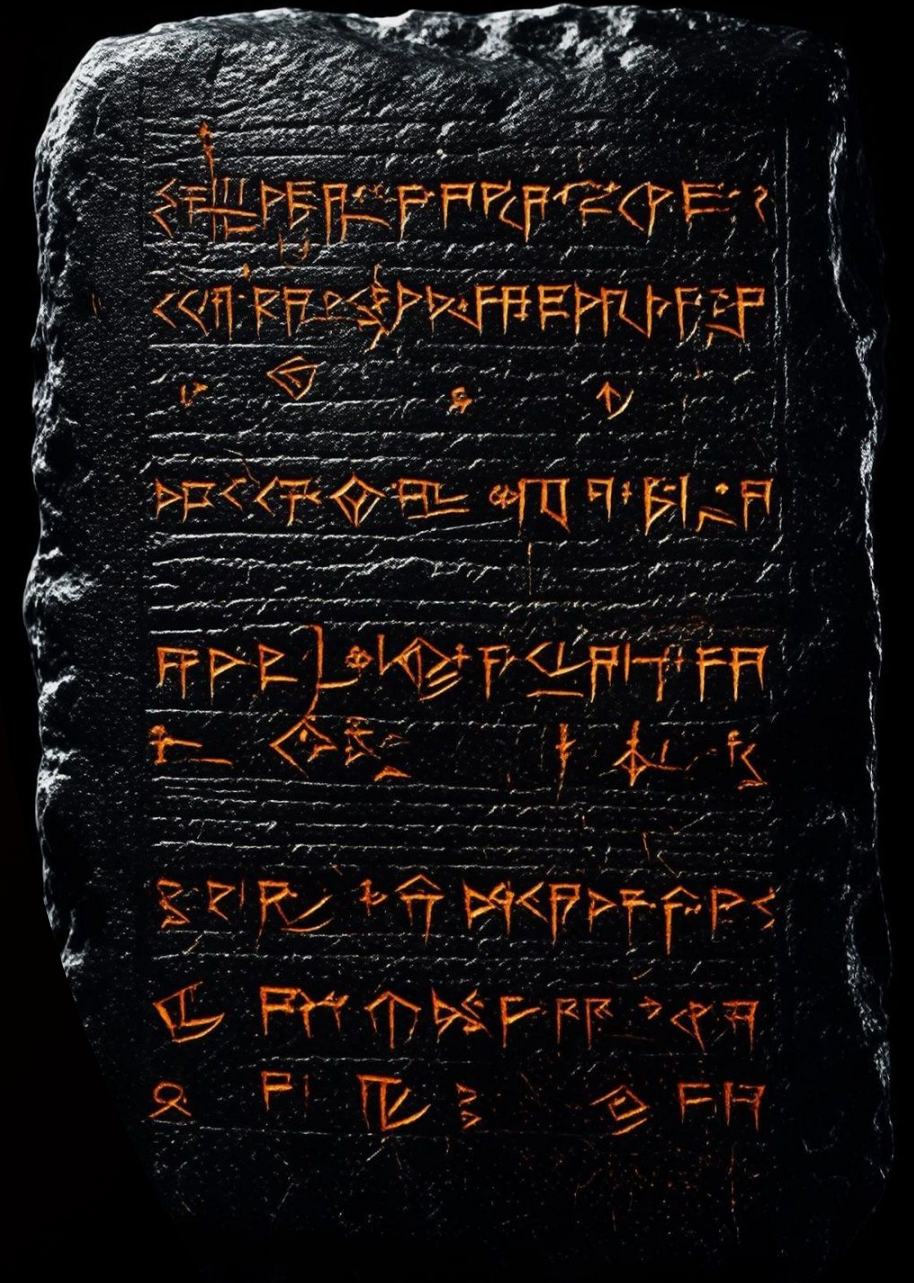
”

“

"Building a tool that can encrypt, decrypt and analyze ciphertext demonstrates both programming and analytical skills relevant to cybersecurity."

— **Industry Relevance**

”



Technology Stack

A modern full-stack architecture connecting an interactive React frontend to Python cryptographic services via FastAPI, with PostgreSQL storing all operation history.



PostgreSQL & SQLAlchemy

Persistent **operation history** storage via database ORM.



FastAPI & Python

Backend REST API serving **cryptographic logic** and analysis services.



React & Vite

Interactive **frontend interface** with HTML/CSS for a seamless user experience.

Caesar Cipher



Encryption

Shift each letter by a fixed number of positions

- **Plaintext:** MY NAME IS JATIN MEHTA
- **Shift:** 3 → Ciphertext: PB QDPH LV MDWLQ PHHWD

Caesar Cipher

Vigenère Cipher

Frequency Analysis

History

Caesar Cipher

Shift letters by a fixed number to encrypt or decrypt your message.

CLASSICAL CIPHER

Input Text

MY NAME IS JATIN MEHTA

Shift Value

3

Encrypt

Decrypt

Brute Force

Result

SUCCESS

PB QDPH LV MDWLQ PHHWD



Decryption

Reverse the shift to recover the original plaintext

- **Key space:** Only 26 possible shifts — exhaustive brute force is practical
- **Scoring:** Results ranked by English-language likelihood

Caesar Cipher

Vigenère Cipher

Frequency Analysis

History

Caesar Cipher

Shift letters by a fixed number to encrypt or decrypt your message.

CLASSICAL CIPHER

Input Text

PB QDPH LV MDWLQ PHHWD

Shift Value

3

Encrypt

Decrypt

Brute Force

Result

SUCCESS

MY NAME IS JATIN MEHTA

Caesar Cipher

Vigenère Cipher

Frequency Analysis

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Caesar Cipher

Shift letters by a fixed number to encrypt or decrypt your message.

CLASSICAL CIPHER

Input Text

PB QDPH LV MDWLQ PHHWD

Shift Value

3

Encrypt

Decrypt

Brute Force

Result

SUCCESS

BEST CANDIDATE

MY NAME IS JATIN MEHTA

SHIFT: 3 Score: 0.2

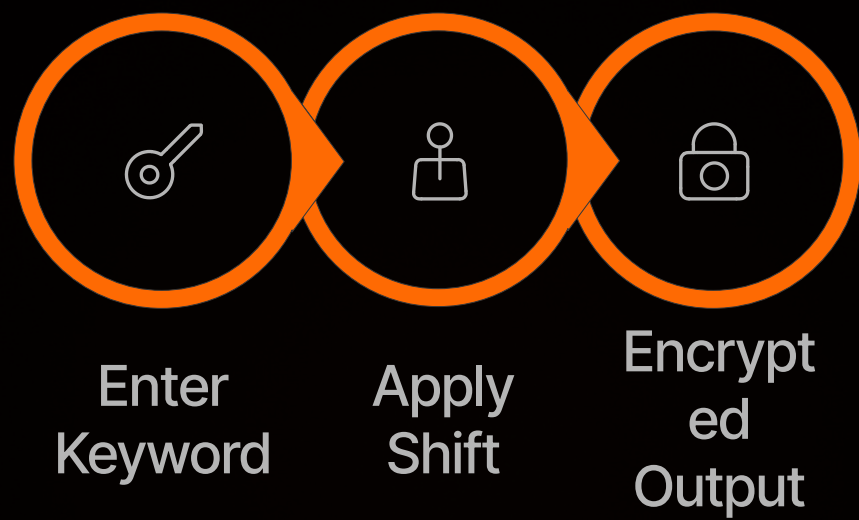


Brute Force

Test all 26 shifts, rank by English-language scoring

Vigenère Cipher

Polyalphabetic substitution — uses a repeating keyword instead of a single fixed shift, making frequency analysis much harder than Caesar.



Example

Plaintext: MY NAME IS JATIN MEHTA
Key: CRYPTO
The same keyword can be applied in reverse to decrypt and recover the original message.

Explore classical encryption techniques and analyze encrypted text using an interactive toolkit.

Caesar CipherVigenère CipherFrequency AnalysisHistory

Vigenère Cipher

Use a repeating secret key to encrypt or decrypt your message.

Input Text

MY NAME IS JATIN MEHTA

Encryption Key

CRYPTO

EncryptDecrypt

Result

OP LPFS KJ HPMWP DQWMO

SUCCESS

Explore classical encryption techniques and analyze encrypted text using an interactive toolkit.

Caesar CipherVigenère CipherFrequency AnalysisHistory

Vigenère Cipher

Use a repeating secret key to encrypt or decrypt your message.

Input Text

OP LPFS KJ HPMWP DQWMO

Encryption Key

CRYPTO

EncryptDecrypt

Result

MY NAME IS JATIN MEHTA

SUCCESS

Frequency Analysis

Letter Occurrence

Counts how often each letter appears in the input text to reveal **distribution patterns**.

Percentage Frequency

Calculates each letter's share of total characters, enabling comparison against **known English frequencies**.

Visual Frequency Bars

Displays results as interactive bars, making cipher patterns **immediately visible** to the analyst.

Frequency Analysis

CRYPTANALYSIS

Measure how frequently each letter appears in the supplied text.

Text to Analyze

MY NAME IS JATIN MEHTA.

Analyze Frequency

Frequency Results

ANALYZED

M Count: 3 16.67%	A Count: 3 16.67%
N Count: 2 11.11%	E Count: 2 11.11%
I Count: 2 11.11%	T Count: 2 11.11%
Y Count: 1 5.56%	S Count: 1 5.56%
J Count: 1 5.56%	H Count: 1 5.56%

Cryptanalysis Approach

The toolkit demonstrates how classical ciphers can be systematically analyzed and attacked using brute-force techniques combined with English-language scoring.

1 Step 1: Generate All Shifts

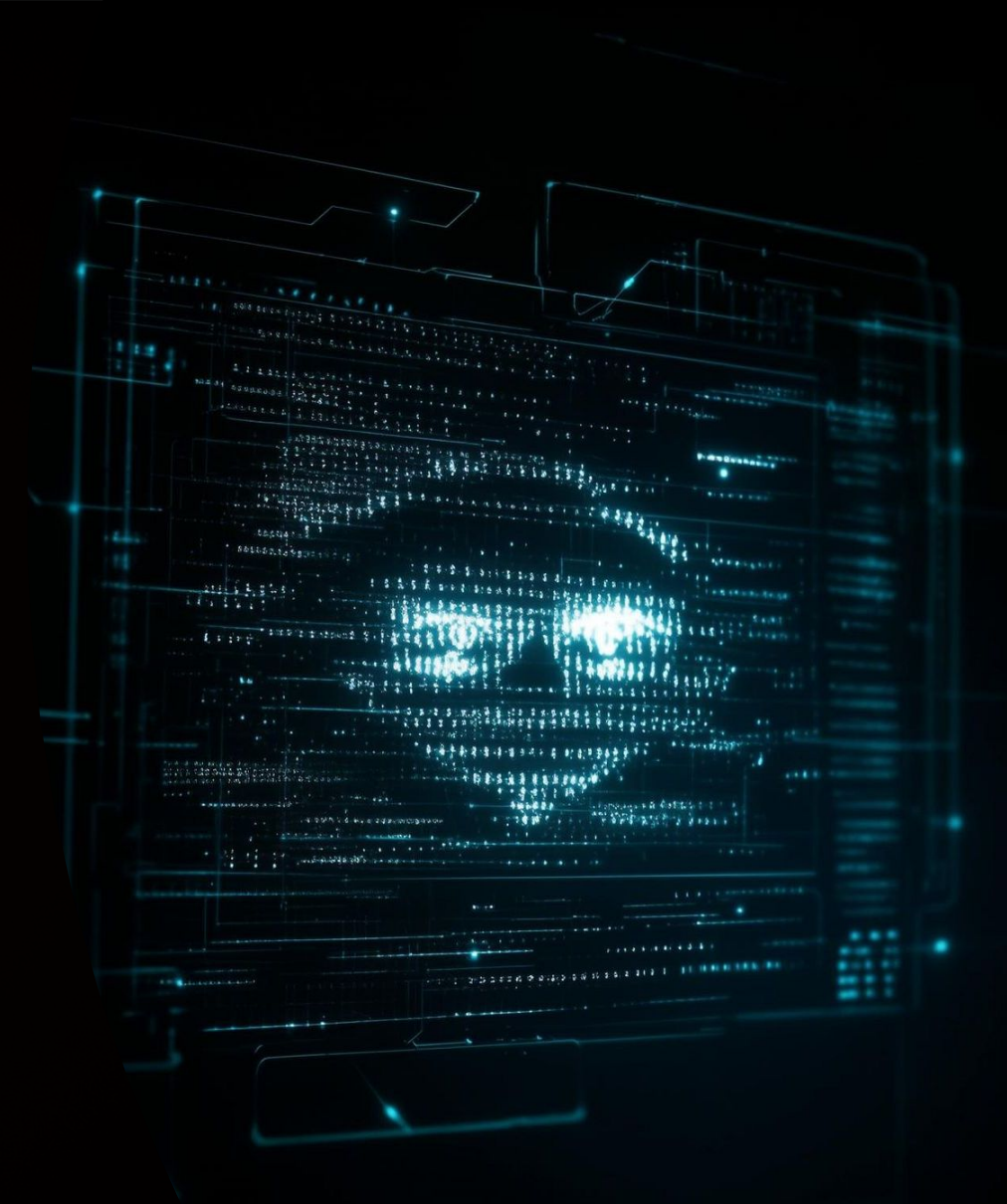
Produce all 26 possible Caesar decryptions from the ciphertext.

2 Step 2: Score Each Candidate

Calculate an English-language score for each decrypted candidate plaintext.

3 Step 3: Rank & Display

Present the best candidate first, with all alternatives available for review.



Database & Operation History

All cryptographic operations are persisted in PostgreSQL, giving users a complete audit trail of their encryption, decryption and analysis work.



cipher_history

Algorithm, operation type, input text, key / shift, output text, timestamp



analysis_history

Input text, letter counts, percentage frequencies, timestamp



Review & Replay

Users can browse all previous operations at any time through the history interface

PostgreSQL with SQLAlchemy ORM ensures reliable, queryable storage of every operation performed in the toolkit.

```
curl -X 'GET' \
  'http://127.0.0.1:8080/history/analysis' \
  -H 'accept: application/json'

Request URL
http://127.0.0.1:8080/history/analysis

Server response
Code 200
Details

Response body
{
  "id": 1,
  "analysis_type": "Frequency Analysis",
  "input_text": "MY NAME IS JATIN MEHTA",
  "result": [
    [
      {"letter": "M", "count": 2, "percentage": 16.67},
      {"letter": "Y", "count": 1, "percentage": 8.33},
      {"letter": "N", "count": 2, "percentage": 16.67},
      {"letter": "A", "count": 1, "percentage": 8.33},
      {"letter": "E", "count": 1, "percentage": 8.33},
      {"letter": "S", "count": 1, "percentage": 8.33},
      {"letter": "I", "count": 1, "percentage": 8.33},
      {"letter": "J", "count": 1, "percentage": 8.33},
      {"letter": "T", "count": 1, "percentage": 8.33},
      {"letter": "H", "count": 1, "percentage": 8.33}
    ]
  ],
  "created_at": "2020-09-18T07:57:09.873482"
}
```

```
curl -X 'GET' \
  'http://127.0.0.1:8080/history/cipher' \
  -H 'accept: application/json'

Request URL
http://127.0.0.1:8080/history/cipher

Server response
Code 200
Details

Response body
{
  "id": 1,
  "algorithm": "Caesar Cipher",
  "operation": "Encryption",
  "input_text": "MY NAME IS JATIN MEHTA",
  "key_or_shift": 2,
  "output_text": "OY OQNI LV KQALQ PHQDU",
  "created_at": "2020-09-18T07:58:10.442994"
}
```

Results & Conclusion

This project successfully brought together cryptography, web development, and data persistence into a complete, working application. It demonstrates how classical cipher techniques can be implemented, tested, and analyzed in a modern software environment. Beyond the technical implementation, the toolkit provides a strong educational foundation for understanding encryption, decryption, and cryptanalysis. It also highlights the practical importance of securing information and studying attack methods in cybersecurity. Overall, the project delivers both a functional solution and a meaningful learning platform for exploring core security concepts.

